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Hybrid Concept Bottleneck Models

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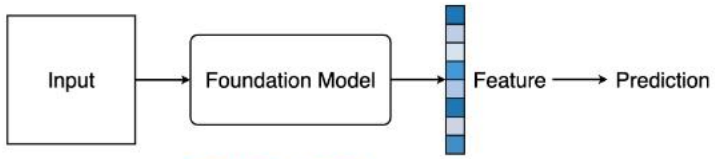
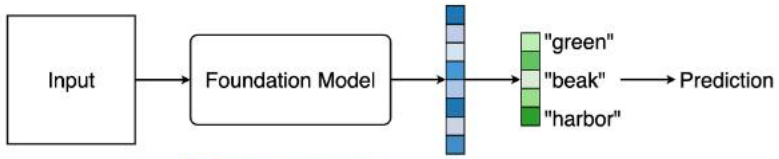
Background

- **黑盒模型的局限：** 传统的深度神经网络（如 ResNet, ViT）虽然准确，但决策过程是“黑盒”，人类无法理解其内部逻辑。
- **CBM 的定义：** 引入一个中间层（概念层），将图像映射为人类可理解的语义概念（如“有翅膀”、“红色头顶”），再由概念预测类别。
- **核心优势：**
  - **可解释性：** 决策过程透明。
  - **可干预性：** 人类可以修正错误的概念来影响最终结果。



Feature-based Prediction Pipeline



Concept-based Prediction Pipeline

Challenge

1. it is perceived to underperform compared to black-box models
2. the requirement for precise annotations for each concept significantly limits their broader application.

LaBo（CVPR2023):

- 概念：LLM 生成 + 子模块选择（固定）
- 残差：无
- 性能：中等
- 可解释性：高 ☑
- 局限：概念库可能不完整 ✕

ResCBM（CVPR2024）:

- 概念：ConceptNet 候选库 + 可学习向量
- 残差：可学习向量（受候选库正则化）
- 性能：中高
- 可解释性：中 ⚠
- 局限：仍受限于候选概念库范围 ✕

Background

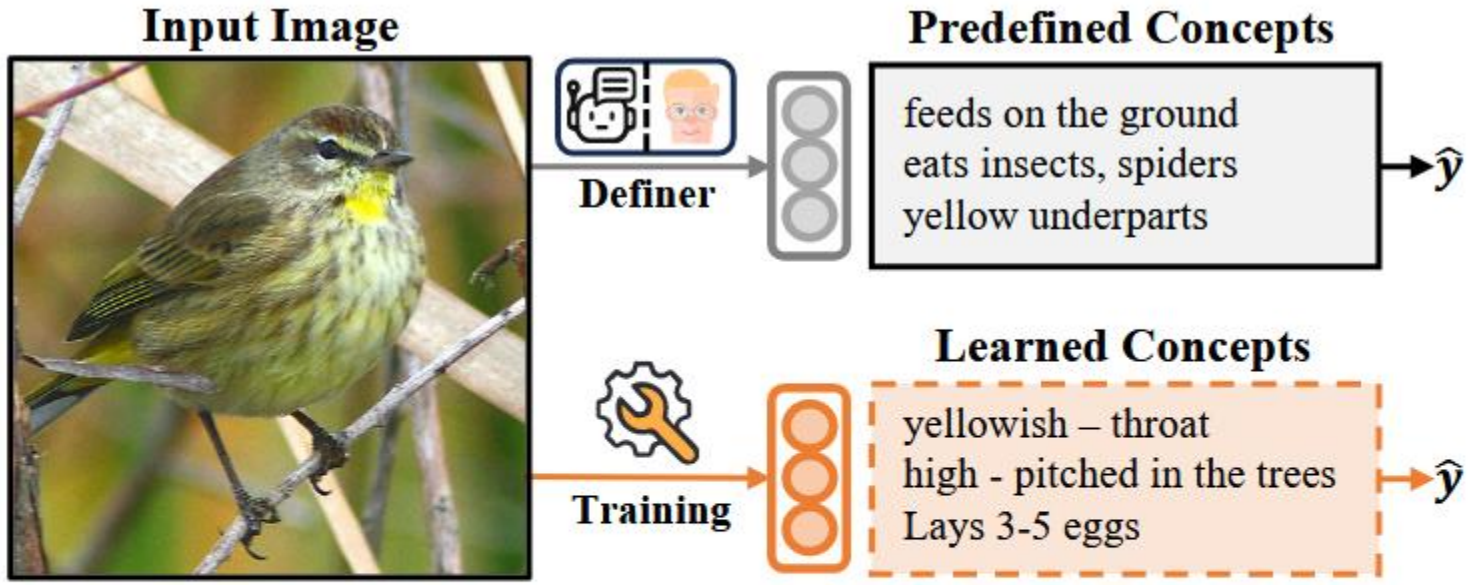


Figure 1. Our proposed concept-trainable CBM reduces the need for meticulously designed concepts by incorporating learned concepts (e.g., 'yellowish throat'), thereby enhancing interpretability and performance while minimizing manual design effort.

Methods

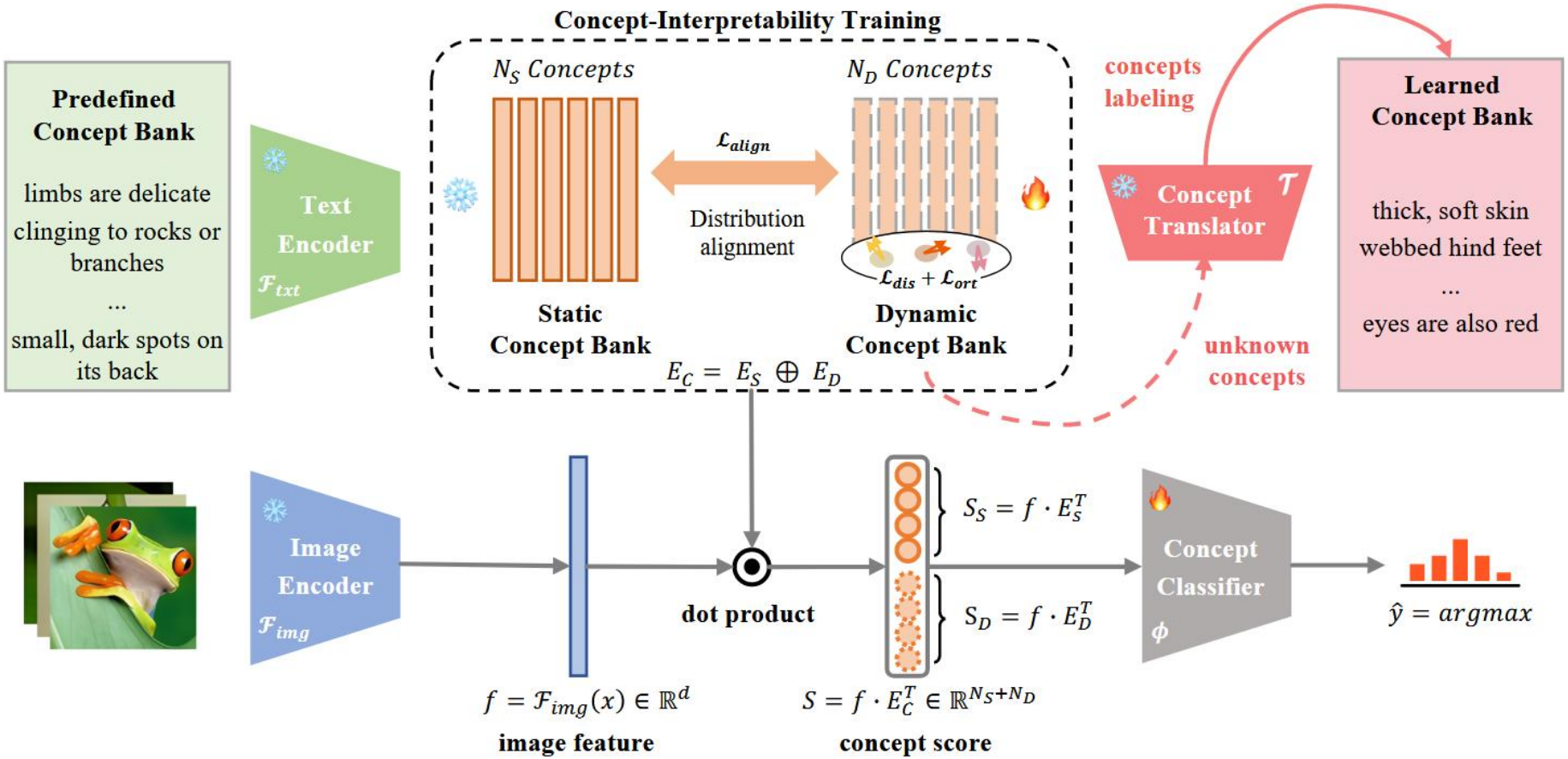


Figure 2. Overview of the **Hybrid** Concept **B**ottleneck **M**odel (**HybridCBM**), which is designed for interpretable concept learning and labeling. First, a Concept Translator is pre-trained to do concept-labeling. Next, a pretrained alignment model (CLIP) is used to obtain embeddings of predefined concepts, forming the Static Concept Bank. The Dynamic Concept Bank learns missing concepts and aligns its distribution with the Static Concept Bank to create the final concept bottleneck. Finally, an interpretable concept-based predictor classifies targets based on the concept scores.

Methods

Static Concept Bank

$$C_S = \{c_1, c_2, \ldots, c_{N_S}\}$$

类别名称 → GPT-3.5生成候选概念 → 概念选择 → CLIP编码 → 静态概念库

↓

500个概念/类

↓

子模块/随机选择

↓

N个概念/类

$$S^* = \arg\max_{|S|=N} F(S)$$

$$F(S) = \underbrace{\text{Coverage}(S)}_{\text{覆盖}} + \lambda_1 \underbrace{\text{Diversity}(S)}_{\text{多样性}} - \lambda_2 \underbrace{\text{Redundancy}(S)}_{\text{冗余}}$$

Methods

Static Concept Bank

$$\text{Coverage}(S) = \sum_{i \in \text{all}} \max_{j \in S} \text{sim}(f_i, f_j)$$

选出的集合尽量“代表全部500个”

$$\text{Diversity}(S) = \sum_{i \neq j \in S} (1 - \text{sim}(f_i, f_j))$$

让概念具有多样性，不重复

$$\text{Redundancy}(S) = \sum_{i \neq j \in S} \text{sim}(f_i, f_j)$$

避免避免重复

$$\Delta(c) = F(S \cup \{c\}) - F(S)$$

如果选这个 c，会带来多少新增价值

Method	1	2	4	8	16	Full	Avg
Linear Probe	52.30	64.60	71.94	78.33	81.55	**86.84**	72.59
Labo	N/A	N/A	N/A	N/A	N/A	85.72	N/A
HybridCBM (submodular)	**58.64**	**69.52**	**75.08**	**78.71**	**81.93**	86.83	**75.12**
HybridCBM (random)	**57.58**	**66.69**	**74.62**	78.02	80.98	86.82	**74.16**

Table 1. Mean accuracy across all datasets, at different shots.

Methods

Dynamic Concept Bank

$$C_D = \{c_1, c_2, \ldots, c_{N_D}\}$$

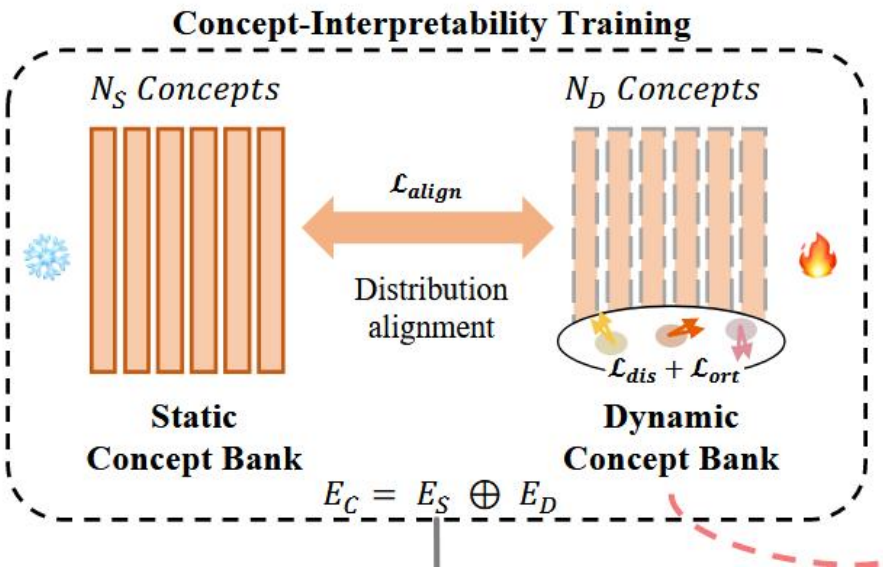
动态概念文本概念空间

$$E_D = \{E_D^1, E_D^2, \ldots, E_D^K\}$$

随机初始化的可学习向量

$$e_d \in \mathbb{R}^d \qquad c = \mathcal{T}(e)$$

目标是获得一组可学习的概念，这些概念具有判别性、多样性和与人类可理解的含义在语义上保持一致，从而提高模型的可解释性和性能。



Methods

Discriminability Loss（判别损失）：

$$\mathcal{L}_{intra}^{i} = y_k \left\{ -\hat{f} \cdot \hat{c}_k + \alpha \left| m - \hat{\mathcal{F}}_{txt}(k) \cdot \hat{c}_k \right| \right\} \quad (2)$$

$$\mathcal{L}_{inter}^{i} = (y_k - 1) \left\{ -\hat{f} \cdot \hat{c}_k + \alpha \left| m - \hat{\mathcal{F}}_{txt}(k) \cdot \hat{c}_k \right| \right\} \quad (3)$$

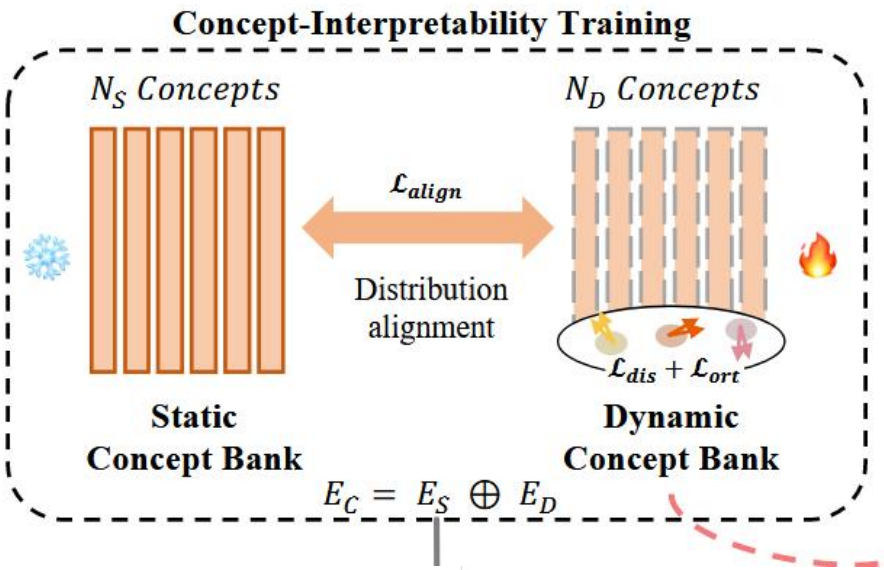
$$\mathcal{L}_{dis} = \mathcal{L}_{intra} + \beta \mathcal{L}_{inter} \qquad {}^{3}\mathbf{c}_k = \mu_k + \sigma_k \cdot \epsilon,$$

Orthogonality Loss（正交损失）：

$$\mathcal{L}_{ort-intra} = \frac{1}{K} \sum_{k=1}^{K} \frac{1}{(N_D/K)^2} \sum_{i \neq j} \left| \hat{\mathbf{e}}_{D,i}^{k} \cdot \hat{\mathbf{e}}_{D,j}^{k} \right| \quad (4)$$

$$\mathcal{L}_{ort-inter} = \frac{1}{N_S N_D} \sum_{i=1}^{N_S} \sum_{j=1}^{N_D} \left| \hat{\mathbf{e}}_{S,i} \cdot \hat{\mathbf{e}}_{D,j} \right| \quad (5)$$

$$\mathcal{L}_{ort} = \mathcal{L}_{ort-intra} + \mathcal{L}_{ort-inter}$$



y_k	类别指示函数
$\hat{f}$	图像特征向量
$\hat{c}_k$	动态概念向量
$\hat{\mathcal{F}}_{txt}(k)$	类别名称嵌入
m	Margin阈值
α	平衡超参数
β	平衡超参数
$\cdot$	点积

Methods

Distribution Alignment Loss：

$$\mathcal{L}_{align} = \mathfrak{S}_{div,\epsilon}(E_D, E_S)$$

Sinkhorn散度（熵正则化版本的最优传输距离）

Sinkhorn 散度：$O(n^2/\varepsilon^2)$ 矩阵迭代，可并行加速

问题：

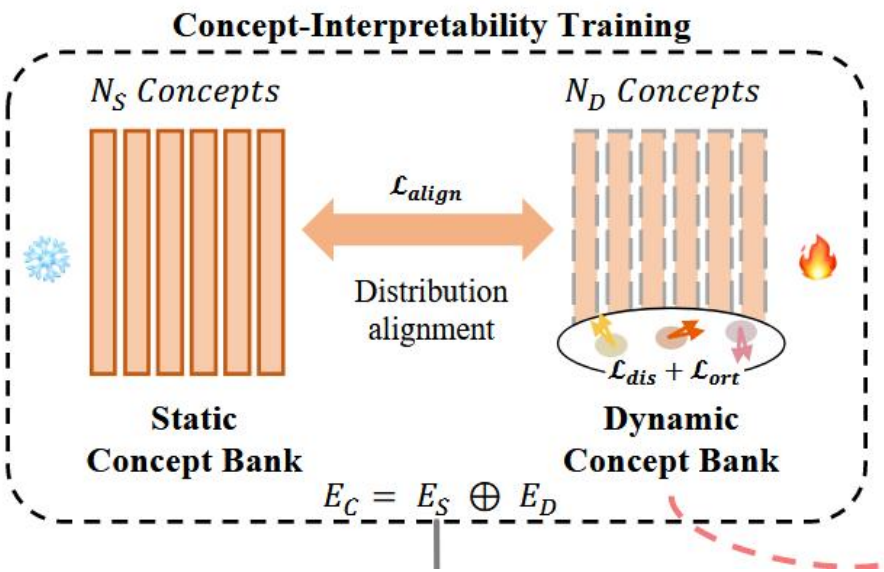
- 动态概念是随机初始化的向量，可能"漂移"到语义空间外
- 可能学到无法解释的纯视觉模式
- Concept Translator难以翻译分布外的向量

解决方案：

- 用Sinkhorn散度将动态概念分布拉回静态概念的分布范围
- 确保动态概念在CLIP语义空间内学习
- 便于翻译器将向量→文本

$$\mathcal{L} = \mathcal{L}_{\text{cls}} + \lambda_{\text{dis}}\mathcal{L}_{\text{dis}} + \lambda_{\text{ort}}\mathcal{L}_{\text{ort}} + \lambda_{\text{align}}\mathcal{L}_{\text{align}}$$

$$\min_{\phi, E_D} \mathop{\mathbb{E}}_{(x,y)\sim\mathcal{D}} \left[\mathcal{L}_{cls}\left(\phi(f \cdot \hat{E_S}^T \oplus f \cdot \hat{E_D}^T), y\right)\right] + \lambda\Omega(\phi) + \Gamma_{E_D}$$



Methods

Concept Translator

GPT-2，12层Transformer，768维隐藏状态

预训练（在HybridCBM训练之前），训练时**冻结**

1,738,985个概念 + 566,747图像-概念对

Suppose an embedding-concept pair (e, c), where $c = \{word_1, word_2, \ldots, word_n\}$, the translator $\mathcal{T}$ learns to reconstruct c conditioned on the preceding words and the l_2-

$$\mathbf{c}_k \in E_D^k \qquad c = \mathcal{T}(e)$$

$$\mathcal{L}_{\text{Recons}} = -\frac{1}{n}\sum_{i=1}^{n} \log P_{\mathcal{T}}(word_i \mid word_{<i}, e) \qquad (10)$$

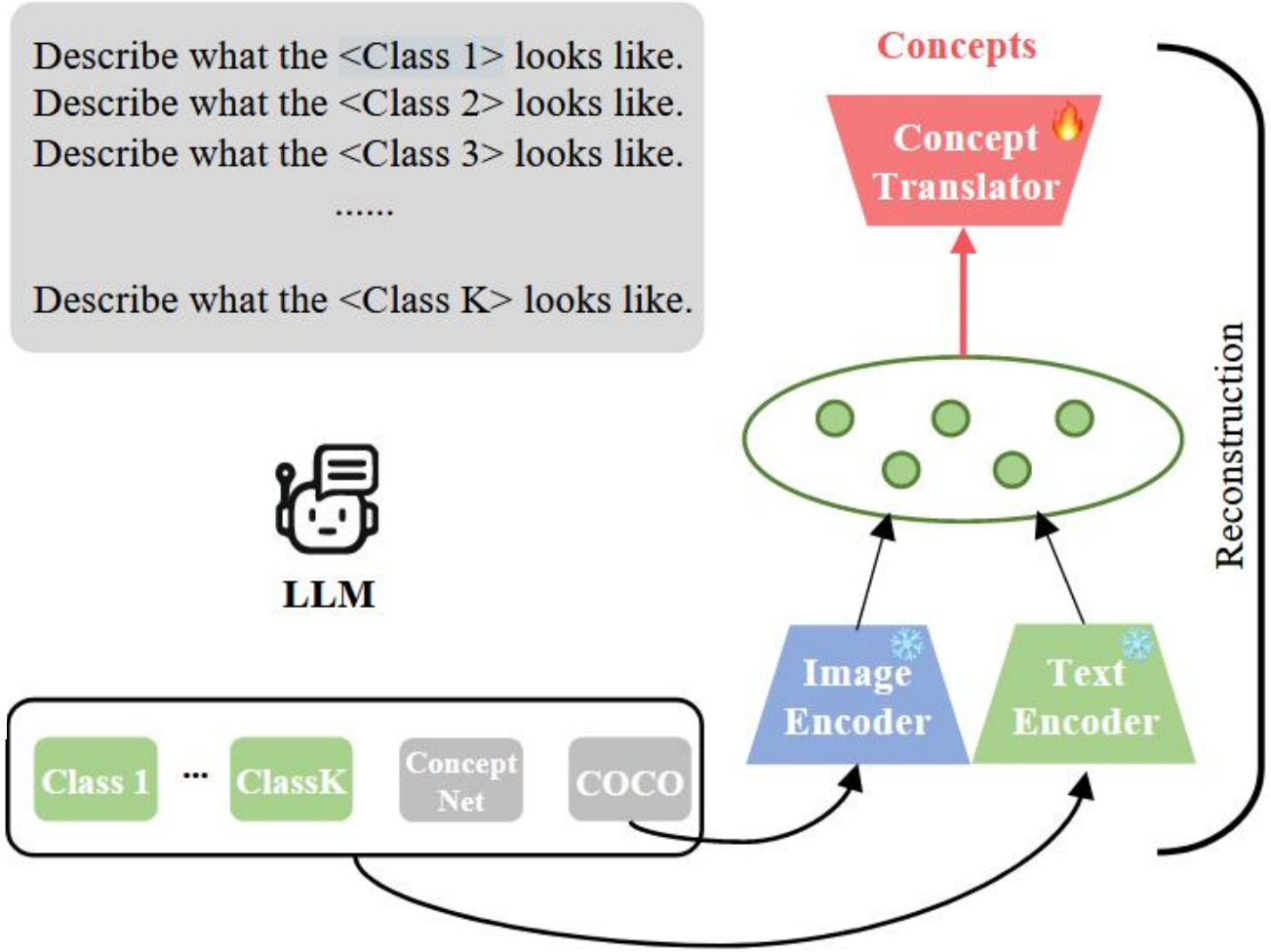


Figure 7. **Concept Translator Pre-training.** The LLM is prompted to generate concepts, which are then combined with COCO captions to train the Concept Translator.

Experiments

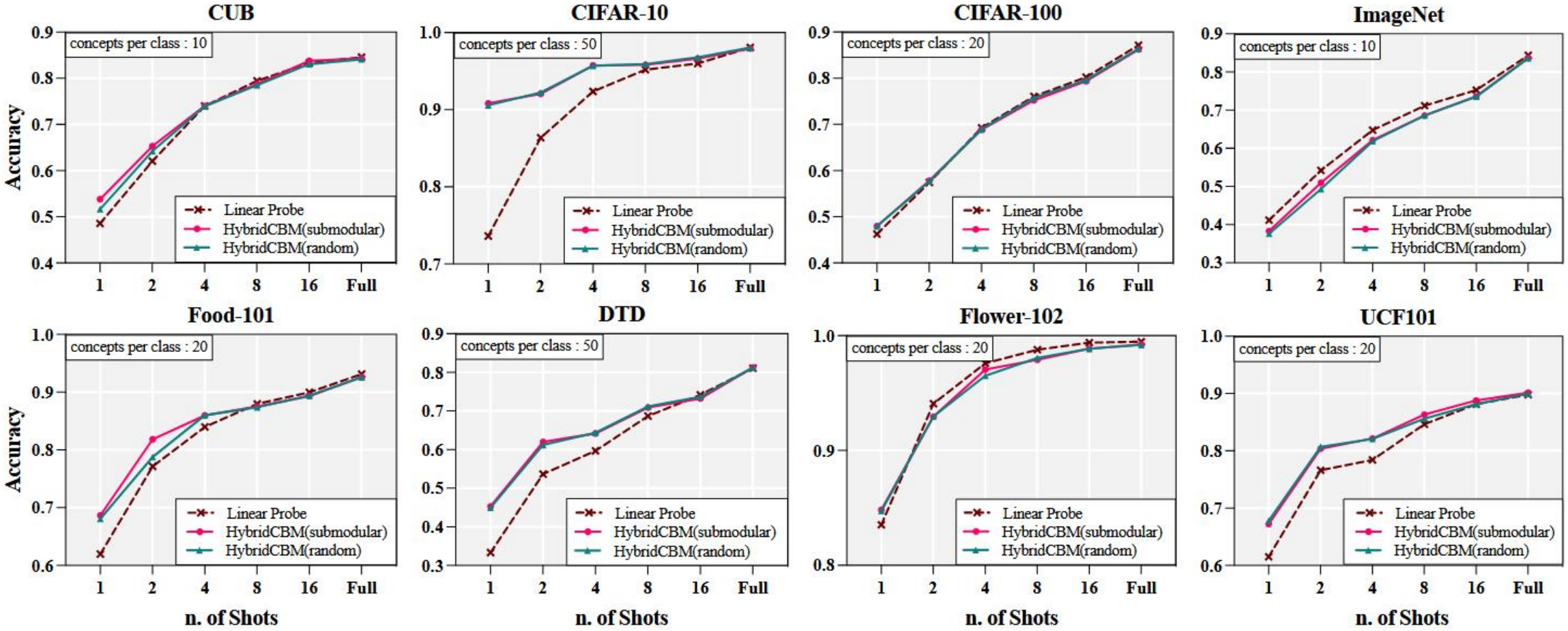


Figure 3. Comparison of test accuracy between HybridCBM with submodular and random selection methods, and Linear Probe across 8 datasets. The x-axis denotes the number of labeled images.

Experiments

$$Purity = \frac{1}{K}\sum_{k=1}^{K}(\bar{\hat{e}}_k \cdot \hat{\mathcal{F}}_{txt}(k)) \quad (6)$$

$$Separation = 1 - \frac{2}{K(K-1)}\sum_{i=1}^{K}\sum_{j=i+1}^{K}(\bar{\hat{e}}_i \cdot \bar{\hat{e}}_j) \quad (7)$$

$$Semantics = \frac{1}{N_D}\sum_{i=1}^{N_D}\delta(LLM(c_i, classes), 1) \quad (8)$$

$$Precision@t = \frac{1}{K}\sum_{i=1}^{K}\frac{\delta(VLM(c_{i,t}, img), 1)}{t} \quad (9)$$

Method	Interpretability	CIFAR-10	CIFAR-100
Linear Probe	✗	88.8	70.1
PCBM-h [54]	✗	87.6	69.9
PCBM [54]	✓	84.5	56.0
Label-Free [26]†	✓	86.3	65.2
Labo [52]	✓	87.6	65.2
ResCBM [39]†	✓	88.0	67.9
CaptionCBM	✓	81.8	60.7
HybridCBM	✓	**88.03**	**68.38**

Table 2. Test accuracy comparison between HybridCBM and on CIFAR-10 and CIFAR-100. † indicates the reported performance.

Loss	Concept Bank	Purity (%)	Separation (%)	Semantics (%)	Precision@t (%)
$\mathcal{L}_{cls}$	Static (submodular)	19.3	86.2	**38.3**	48.5
	Static (random)	19.2	82.4	30.9	47.3
	Dynamic	0.1	100.0	0.01	0.05
$\mathcal{L}_{cls+dis}$	Dynamic	71.9	76.9	30.0	34.8
$\mathcal{L}_{cls+dis+ort}$	Dynamic	72.0	77.1	28.9	34.1
$\mathcal{L}_{cls+dis+ort+align}$	Dynamic	39.8	80.0	**32.8**	**46.2**

Table 3. Evaluation of HybridCBM's static and dynamic concepts across various metrics.

Experiments

	Class Name	Sample	Case interpretability	
			static concepts	dynamic concepts
CUB	Laysan Albatross		1.lays 4-6 white eggs 2.black and white striped body	1.black and white albatross 2.largest albatross species
Food-101	Baklava		1.flaky, phyllo dough texture 2.filled with nuts and sweetened with syrup	1.light brown center 2.a food is arranged on a large platter
Flower	Clematis		1.popular choice for making garlands and wreaths 2.borne on a climbing vine	1.large , a seed plant 2.violet

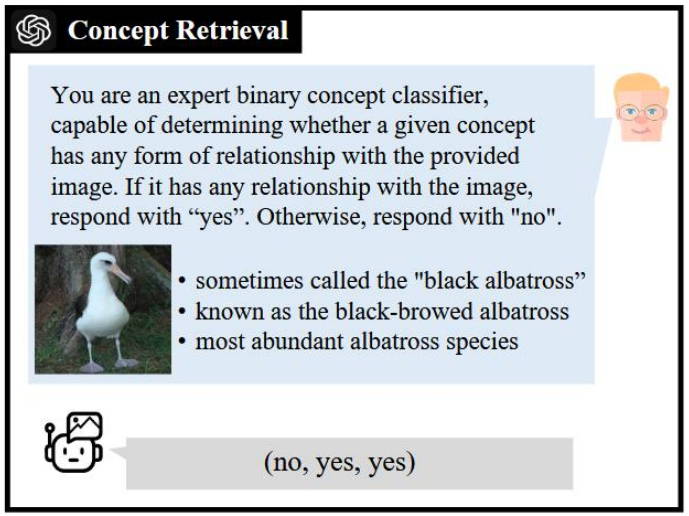


Figure 5. On the left, the top-2 static and dynamic concepts for randomly selected classes across three datasets are presented, focusing on case interpretability. The concept retrieval process with vision-language model GPT-4o is illustrated on the right.

Method	concepts per class	Ratio of Dynamic Concepts					
		0	0.2	0.4	0.6	0.8	1
LP	N/A	**87.09**	N/A	N/A	N/A	N/A	N/A
Hybrid (submodular)	5	83.21	84.46	85.45	86.08	86.34	86.78
	10	84.27	85.47	86.28	86.51	86.87	87.13
	15	84.75	85.84	86.45	86.78	87.08	87.25
	20	84.93	86.09	86.44	86.84	86.88	87.24
Hybrid (random)	5	80.95	84.04	85.28	85.96	86.07	86.53
	10	82.88	85.28	86.30	86.62	86.82	87.13
	15	83.65	85.81	86.54	86.80	86.89	87.25
	20	83.85	86.09	86.66	86.86	87.06	87.24

Table 4. Ablation average results on 10 datasets for varying the ratio of dynamic concepts under different methods.

Method	Metric	$\mathcal{L}_{cls}$	$\mathcal{L}_{cls+dis}$	$\mathcal{L}_{cls+dis+ort}$	$\mathcal{L}_{cls+dis+ort+align}$
Hybrid (submodular)	accuracy	84.60	84.36	84.29	84.54
	precision@t	0.047	34.8	34.1	46.2
Hybrid (random)	accuracy	84.50	84.16	83.91	84.17
	precision@t	0.044	34.8	35.5	47.2

Table 5. Ablation results for varying training loss with a dynamic concept ratio of 0.5 and 10 concepts per class.

Thanks